AI for Logistics and Supply Chains

Practice Week 7
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25
Total Suppliers

$10.39M
Expected Annual Loss

10
Critical Risk Suppliers

$39.04M
Value at Risk (95%)

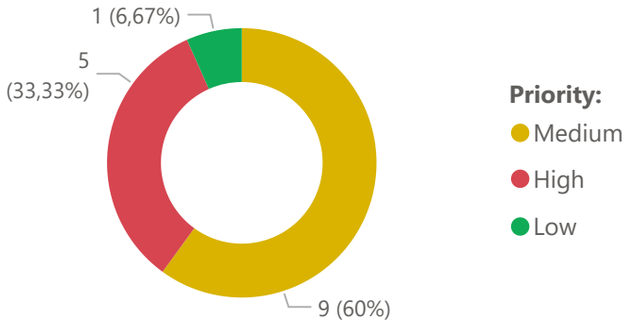
Choose Category:

Cyber	Market
External	Natural
Financial	Quality
Geopolitical	Regulatory
Health	Supplier
Labour	Supply
Logistics	Technology

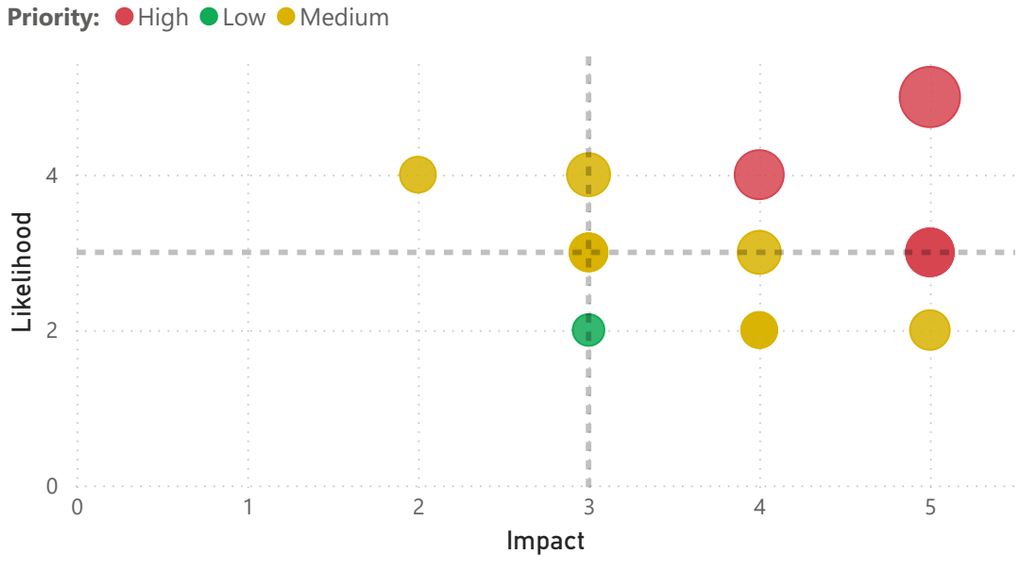
Top 5 Risks

Risk	Risk Score	Control
Semiconductor Shortage	25	Strategic stock
Port Strike (Europe)	16	Dual sourcing
Geopolitical Tension (Taiwan)	15	Diversification
Quality Failure (Critical Part)	15	QMS
Typhoon (SE Asia)	15	Insurance
Всего	**86**	

Risk Portfolio by Priority



Risk Impact vs Likelihood Matrix



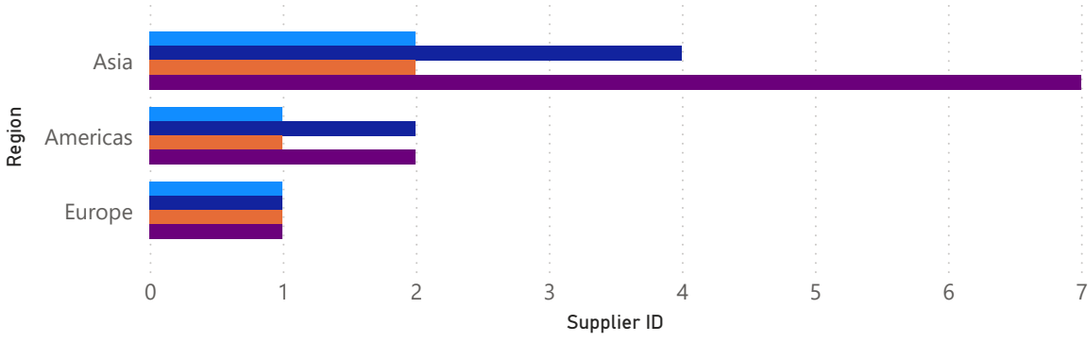
Быстрая мера

Global Spend & Risk Geographic Overview

Risk Tier: Tier 1 - Low Risk · Tier 2 - Medium Risk · Tier 3 - High Risk · Tier 4 - Critical Risk

Supplier Count by Region & Risk Tier

Risk Tier Tier 1 - Low Risk · Tier 2 - Medium Risk · Tier 3 - High Risk · Tier 4 - Critical Risk



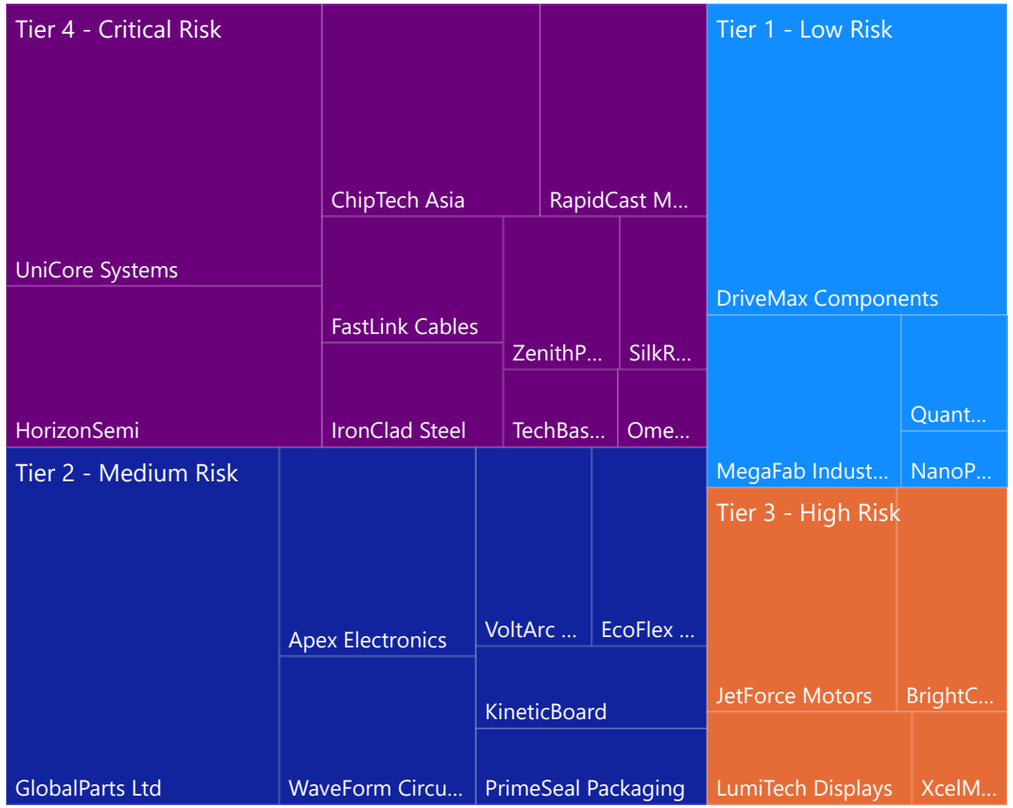
Choose Region:

Americas | Asia | Europe

Choose Risk Tier:

Tier 1 - Low Risk | Tier 2 - Medium Risk | Tier 3 - High Risk | Tier 4 - Critical Risk

Spend Exposure by Supplier & Risk Tier



Single Source:

False

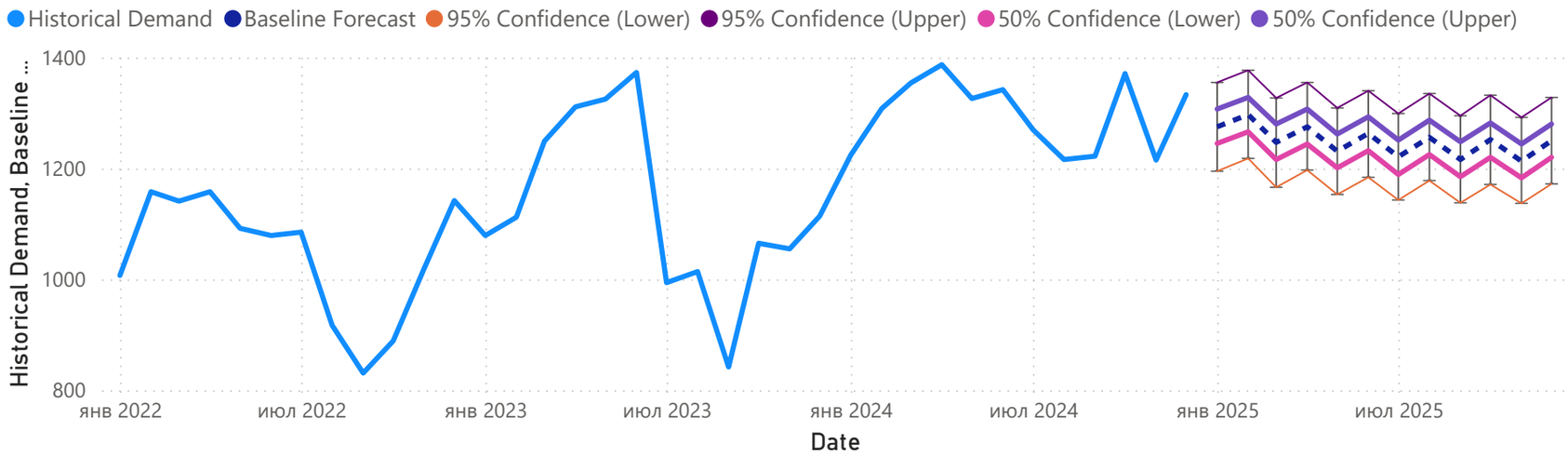
True

High Risk Supplier Portfolio

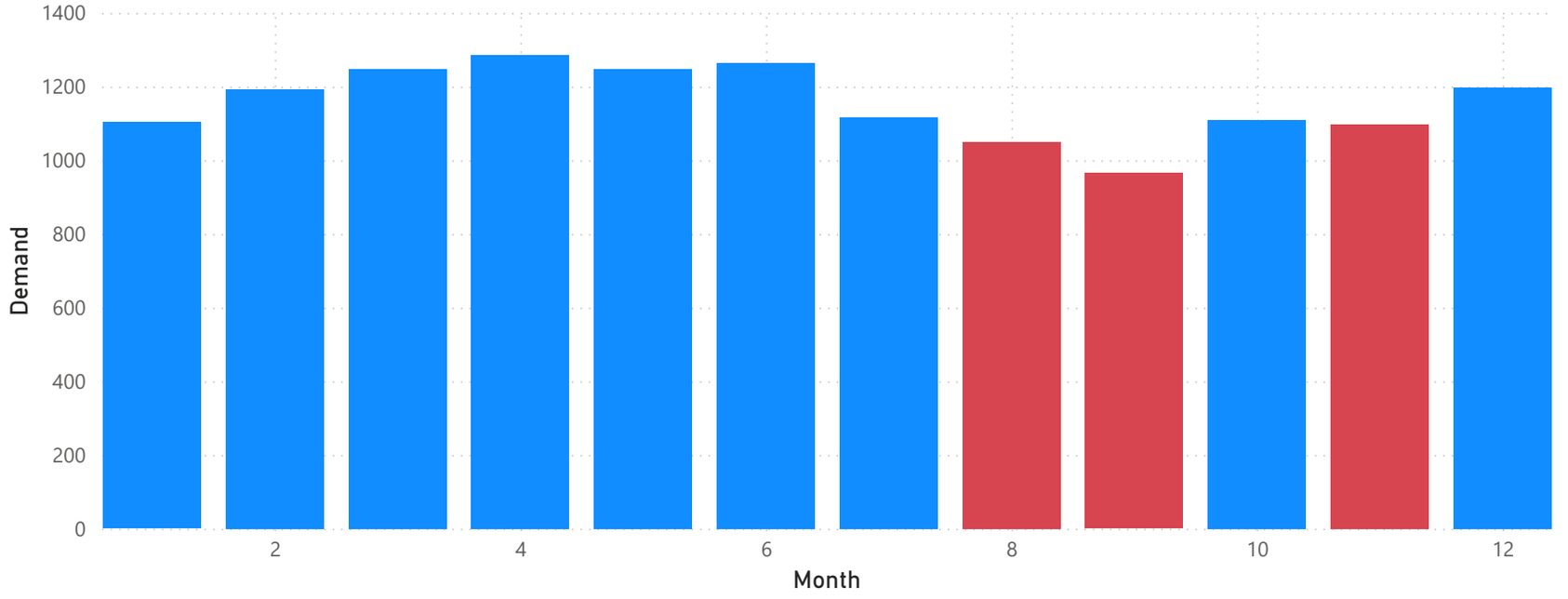
Supplier Name	Annual Spend ($M)	Risk Score	Risk Tier
TechBase Korea	2,39	40,31	Tier 4 - Critical Risk
ZenithParts	4,78	40,07	Tier 4 - Critical Risk
Apex Electronics	10,99	39,99	Tier 2 - Medium Risk
XcelMold	2,36	38,00	Tier 3 - High Risk
UniCore Systems	23,87	37,31	Tier 4 - Critical Risk
RapidCast Metals	9,50	34,08	Tier 4 - Critical Risk
HorizonSemi	13,65	33,65	Tier 4 - Critical Risk
SilkRoute Textiles	3,57	33,48	Tier 4 - Critical Risk
WaveForm Circuits	7,80	33,44	Tier 2 - Medium Risk
Всего	**91,32**		

Быстрая мера

Demand History & 12-Month Uncertainty Forecast



Average Demand by Month (Seasonality Pattern)



Mean Forecast

1,25 тыс.

Expected Monthly Demand

Uncertainty Range

1,33 тыс.

Forecast Uncertainty (P95-P5)

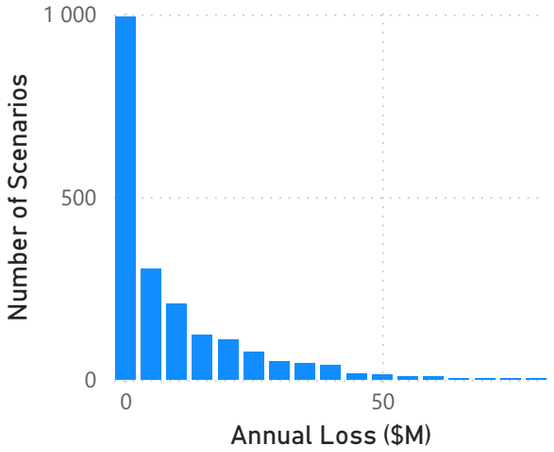
Jan 2025 Target

1,28 тыс.

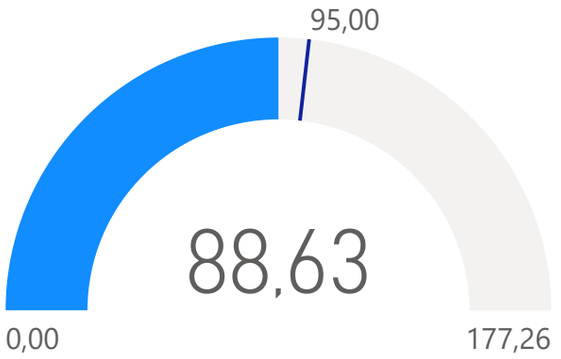
Target Units

Быстрая мера

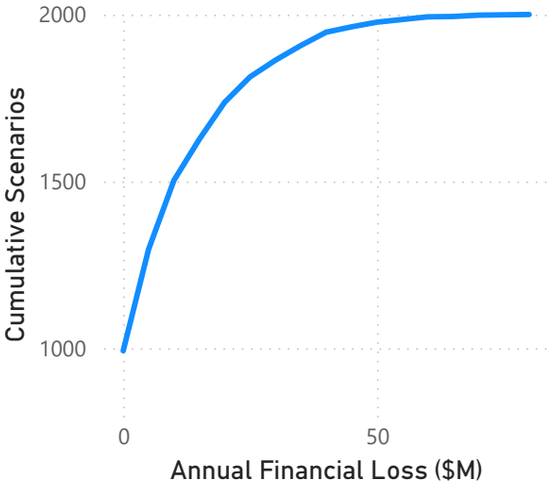
Annual Financial Loss Distribution (10,000 Scenarios)



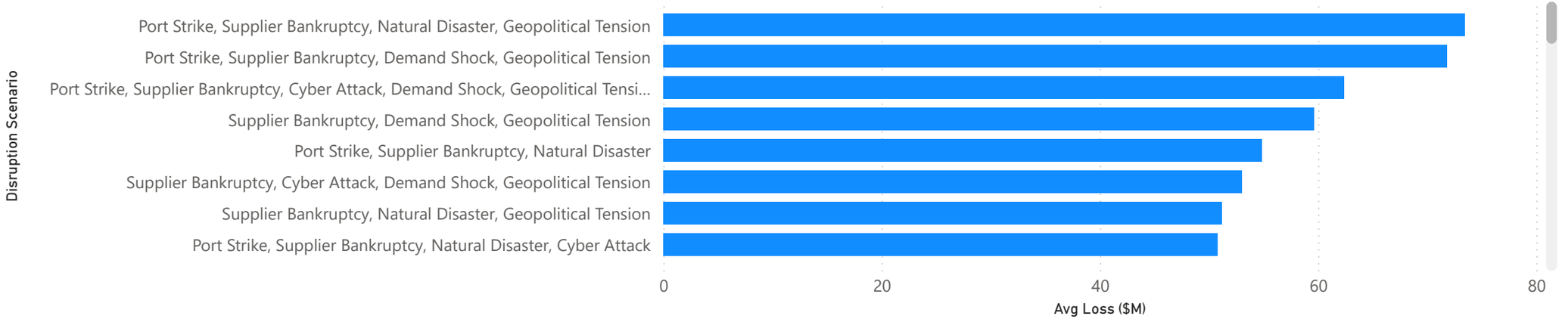
Mean Service Level vs 95% Target



Cumulative Loss Probability (CDF)



Average Impact by Disruption Event



High-Impact Scenarios (>$10M)

704

Total Scenarios

Быстрая мера